THE LEVEL SETS OF THE FIBRE-DIMENSION FUNCTION ARE LOCALLY CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8453-8464 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 8466-8489, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Fibre dimension of a morphism, dimensions of finitely generated algebras, and constructibility; the encoding is retained.

Source reference. Original lemma lemma-base-change-dimension-fibres: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8435–8446. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-dimension-in-neighbourhood: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8412–8417. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8489–8489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 053Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8489–8489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $n_{X/Y}$ be the function on Y giving the dimension of fibres of f introduced in Lemma lemma-base-change-dimension-fibres. Assume f of finite presentation. Then the level sets*

$$E_n = \{y \in Y \mid n_{X/Y}(y) = n\}$$

of $n_{X/Y}$ are locally constructible in Y.

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Proof. Fix n. Let $y \in Y$. We have to show that there exists an open neighbourhood V of y in Y such that $E_n \cap V$ is constructible in V. Thus we may assume that Y is affine. Write $Y = \mathrm{Spec}(A)$ and $A = \mathrm{colim}\, A_i$ as a directed limit of finite type $\mathbf{Z}$-algebras. By Limits, Lemma 01ZM we can find an i and a morphism $f_i : X_i \to \mathrm{Spec}(A_i)$ of finite presentation whose base change to Y recovers f. By Lemma lemma-base-change-dimension-fibres it suffices to prove the lemma for f_i. Thus we reduce to the case where Y is the spectrum of a Noetherian ring.

We will use the criterion of Topology, Lemma 053Z to prove that E_n is constructible in case Y is a Noetherian scheme. To see this let $Z \subset Y$ be an irreducible closed subscheme. We have to show that $E_n \cap Z$ either contains a nonempty open subset or is not dense in Z. Let $\xi \in Z$ be the generic point. Then Lemma lemma-dimension-in-neighbourhood shows that $n_{X/Y}$ is constant in a neighbourhood of ξ in Z. This implies what we want. $\square$